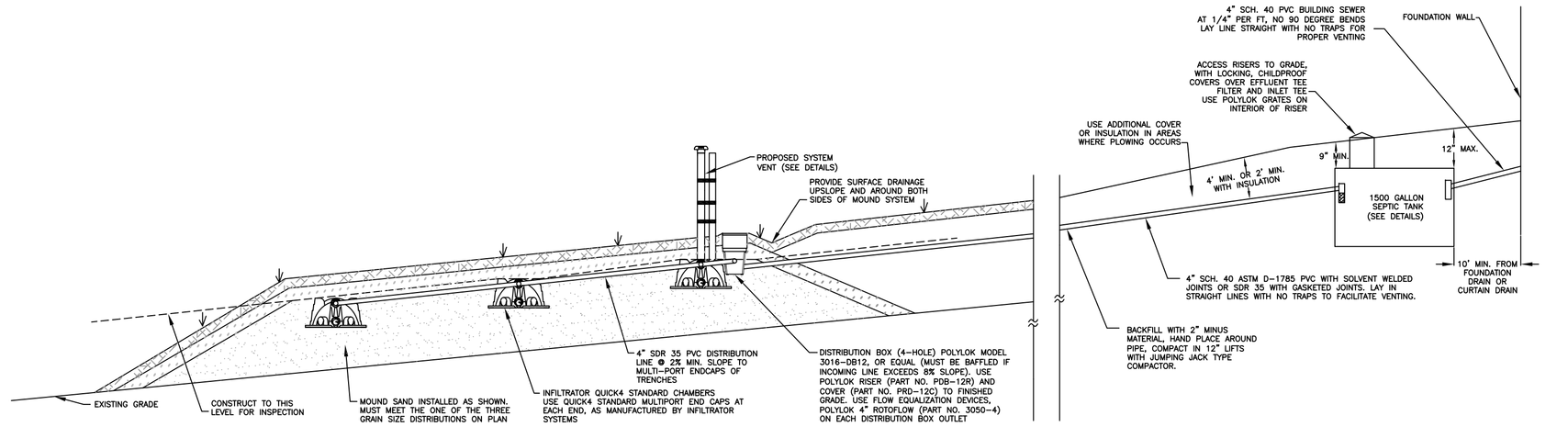
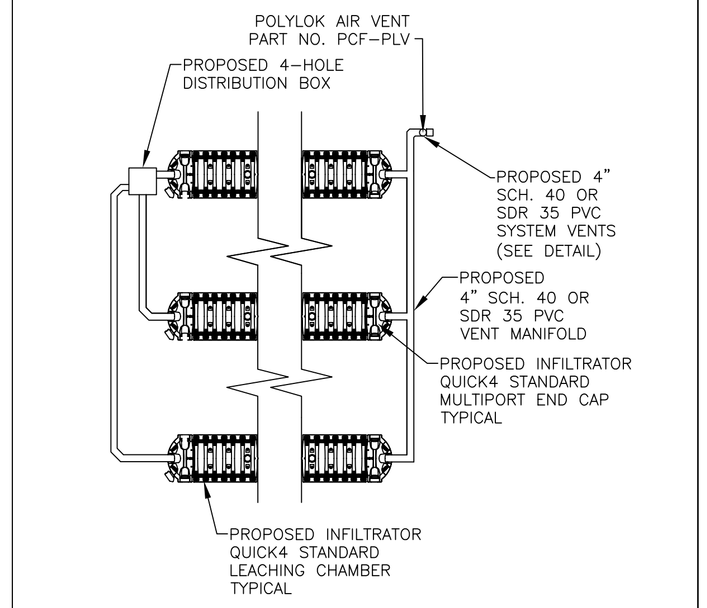


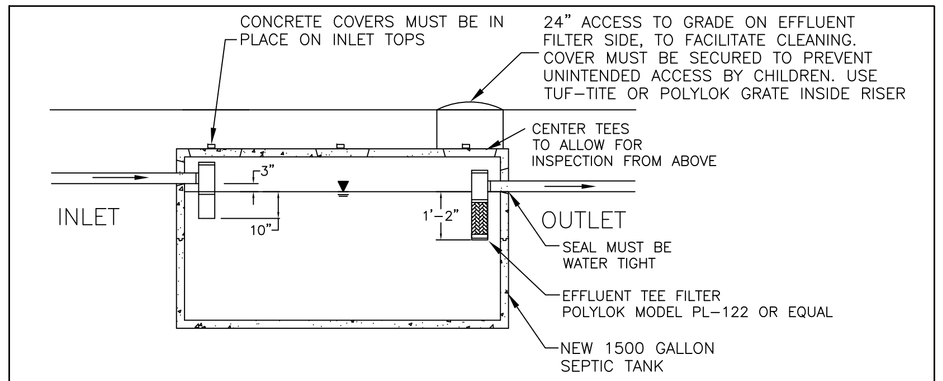
PROFILE THROUGH SYSTEM CENTERLINE

NOT TO SCALE



DISTRIBUTION PIPING & VENT MANIFOLD DETAIL

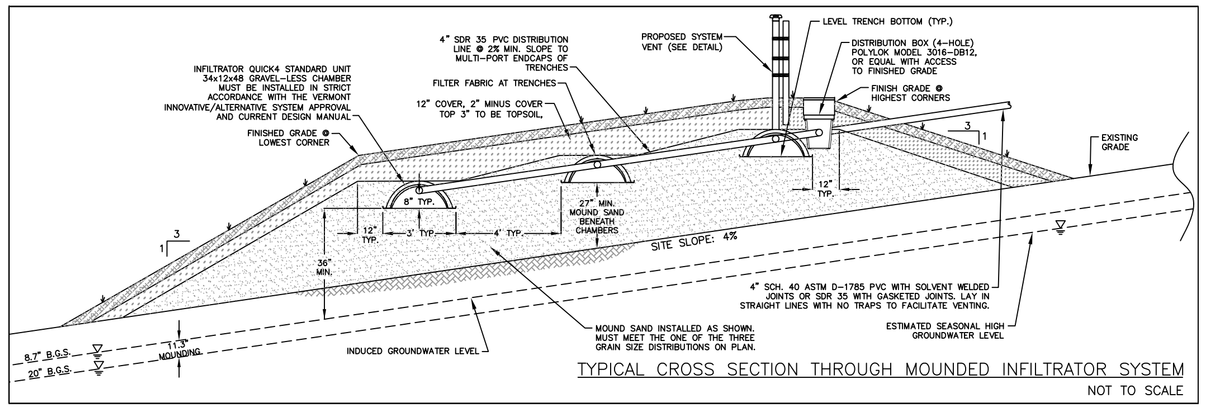
NOT TO SCALE



1500 GALLON SEPTIC TANK DETAIL

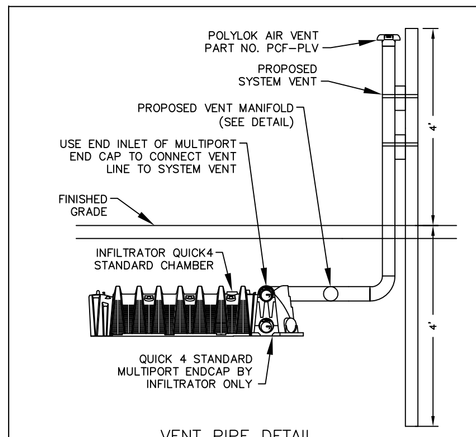
NOT TO SCALE

NOTES:
1) SEPTIC TANK TO BE PRECAST CONCRETE BY WASHBURN VAULT, OR EQUAL. NO WEEP HOLES ALLOWED IN TANK.
2) TANK TO BE SET LEVEL WITH INLET AND OUTLET PENETRATION AS SHOWN.
3) INLET TO USE 4 x 4 x 4 PVC TEE EXTENDING 10" BELOW OUTLET AS SHOWN.
4) OUTLET TO USE 4 x 4 x 4 PVC TEE EXTENDING 14" BELOW OUTLET AS SHOWN.
5) PROVIDE CENTER ACCESS WITHIN 6" OF SURFACE, WITH STEEL PLATE FOR MAGNETIC LOCATOR.
6) ALL PENETRATIONS TO BE SEALED TO ENSURE TANK IS WATERTIGHT.
7) BACKFILL IN 12" LIFTS, WITH COMPACTION USING 4" MINUS MATERIAL.
8) PLACE 6" OF STONE UNDER SEPTIC TANK TO PROVIDE A SOLID BASE.
9) RISERS TO BE HDPE/PLASTIC.



TYPICAL CROSS SECTION THROUGH MOUNDED INFILTRATOR SYSTEM

NOT TO SCALE



VENT PIPE DETAIL

NOT TO SCALE

NOTES:
1) USE POLYLOK AIR VENT PART NO. PCF-PLV AT TOP.
2) USE A 45 ELBOW AT EACH END PIPE, TO CONNECT TO VENT HEADER.
3) USE 4x4x8' P.T. POST WITH STAINLESS STEEL STRAPS TO ATTACH POST TO VENT.
4) IF POSSIBLE USE SDR 35 SOLID WITH STREET ELBOWS, OR 4" SCH 40 PVC.
5) ALL PIPING MUST BE THE SAME COLOR.

SYSTEM VENT DETAIL

NOT TO SCALE

DESIGN LIMITATIONS AND OPERATIONAL NOTES:
1) SYSTEM IS NOT DESIGNED TO SUPPORT A GARBAGE DISPOSAL AND NONE SHALL BE INSTALLED.
2) SYSTEM IS NOT DESIGNED TO HANDLE BACKWASH FROM A WATER TREATMENT DEVICE AND SHALL NOT BE CONNECTED INTO WASTEWATER DISPOSAL SYSTEM.
3) PERIMETER DRAINAGE OR SUMP PUMPS FROM EXISTING RESIDENCE SHALL NOT BE CONNECTED INTO WASTEWATER DISPOSAL SYSTEM AND MUST NOT INFILTRATE OR DISCHARGE WITHIN 50' OF MOUND SYSTEM.
4) THIS DESIGN IS INTENDED TO TREAT NORMAL RESIDENTIAL WASTEWATER. IT IS NOT DESIGNED TO TREAT WASTEWATER CONTAINING SALTS, SOLVENTS OR HAZARDOUS CHEMICALS. THE USE OF CHEMOTHERAPEUTIC OR HEAVY PRESCRIPTION DRUGS CAN HAVE SERIOUS DETRIMENTAL IMPACTS ON THE LIFE OF THE SYSTEM.
5) EXCESSIVE FLOWS FROM LEAKING FIXTURES, RISERS, TANKS, OR OVER OCCUPANCY INVALIDATE THIS DESIGN.
6) NO PRIVATE WATER SUPPLY WELL OR SUCTION LINE SHALL BE LOCATED WITHIN 50 FEET OF THE SEPTIC TANK OR WITHIN 100 FEET OF THE WASTEWATER DISPOSAL SYSTEM.
7) FINISHED GRADE ABOVE THE LEACHING AREA SHALL HAVE A MINIMUM SLOPE OF 2% AND ALL SITE GRADING IS TO BE DONE IN A MANNER THAT PREVENTS THE COLLECTION OF SURFACE WATER.
8) EFFLUENT FILTER TO BE INSTALLED IN SEPTIC TANK OUTLET TEE, WHICH MUST REMAIN IN PLACE AND BE MAINTAINED AS REQUIRED TO MAINTAIN FLOW AND VENTING.
9) EFFLUENT FILTER TO BE MAINTAINED AS PER MANUFACTURER SPECIFICATIONS. THE FILTER WILL CLOG IF NOT PERIODICALLY CLEANED.

INVERT AND SURFACE ELEVATIONS:

FOUNDATION WALL[2]	1962.17	USE 1/4"/FT MIN.
SEPTIC TANK INLET[2]	1961.79	
SEPTIC TANK OUTLET[2]	1961.54	
DISTRIBUTION BOX INLET	1960.68	
DISTIBUTION BOX OUTLET	1960.58	
INVERT OF 4" INLET TO MULTIPORT ENDCAP AT UPPER TRENCH	1960.38	
INVERT OF 4" INLET TO MULTIPORT ENDCAP AT SECOND TRENCH	1959.75	
INVERT OF 4" INLET TO MULTIPORT ENDCAP AT LOWER TRENCH	1959.12	
ELEV. OF HIGHEST TRENCH BOTTOM	1959.72	
ELEV. OF SECOND TRENCH BOTTOM	1959.09	
ELEV. OF LOWEST TRENCH BOTTOM	1958.46	
FINISHED GRADE @ HIGHEST CORNERS	1961.94	
FINISHED GRADE @ LOWEST CORNERS	1960.23	

NOTES:
1) CONTRACTOR TO VERIFY ELEVATIONS PRIOR TO CONSTRUCTION.
2) MINIMUM ELEVATIONS REQUIRED FOR GRAVITY PITCH TO DISTRIBUTION BOX.

FINAL INSPECTION REQUIREMENTS AND PROCEDURE:

1) THE COST FOR ONE SITE VISIT TO COMPLETE THE FINAL INSPECTION HAS BEEN BILLED TO THE CLIENT. THE COST FOR PARTIAL OR INCOMPLETE FINAL INSPECTIONS MAY BE CHARGED DIRECTLY TO THE CONTRACTOR.
2) CONTRACTOR MUST BE PRESENT AT THE TIME OF INSPECTION AND MUST CONTACT THE LOCAL BOARD OF HEALTH OR SEPTIC OFFICER TO COORDINATE THEIR INSPECTION REQUIREMENTS.
3) THE ENTIRE SYSTEM MUST BE SUBSTANTIALLY CONSTRUCTED AND COMPLETELY TESTED BY THE CONTRACTOR PRIOR TO REQUESTING THE FINAL INSPECTION BY THE ENGINEER.
4) SUBSTANTIALLY CONSTRUCTED MEANS THE FOLLOWING:
   FOR ALL TYPES OF SYSTEMS:
   - A. ALL COMPONENTS IN THE LEACH FIELD MUST BE INSTALLED, CONNECTED AND PROPERLY BEDDED TO THE TOP OF CHAMBERS AS SHOWN ON PROFILE.
   - B. THE BUILDING SEWER MUST BE INSTALLED AND BEDDED AND AT LEAST PARTIALLY VISIBLE TO ALLOW INSPECTION OF THE CLASS OF PIPE AND BEDDING MATERIALS.
   - C. THE SEPTIC TANK MUST BE INSTALLED AND BACKFILLED TO THE HEIGHT OF THE BOTTOM OF THE INLET AND OUTLET PIPES TO ALLOW INSPECTION OF THE PENETRATION SEALS. THE INLET TEE, OUTLET TEE, EFFLUENT TEE FILTER MUST BE INSTALLED. ALL RISERS MUST BE COMPLETELY INSTALLED AND AT FINISHED GRADE. ALL PENETRATIONS MUST BE SEALED.
   - D. IF THE DESIGN CALLS FOR A PUMP CHAMBER, IT MUST BE INSTALLED AND BACKFILLED TO THE LEVEL OF THE BOTTOM OF THE PIPE PENETRATIONS. ALL PUMPS, PIPES, SEALS, CABLES, FLOATS, JUNCTION BOXES, ALARMS, RISERS AND COVERS MUST BE FULLY INSTALLED, AND COMPLETELY OPERATIONAL. PROVISIONS TO TEST THE PUMP SYSTEM WITH WATER WILL BE REQUIRED. CONTRACTOR MUST HAVE WATER LEVEL AT THE "PUMP OFF" LEVEL PRIOR TO INSPECTION, AND MUST HAVE SUFFICIENT WATER AVAILABLE TO TEST PUMP AS MANY TIME AS WILL BE NECESSARY.
   - E. THE CONTRACTOR MUST PROVIDE TIE DISTANCES TO ALL COMPONENTS INCLUDING, BUILDING SEWERS, BENDS, TANKS, PUMP CHAMBERS, DISTRIBUTION BOXES AND FIELD CORNERS. THESE TIE DISTANCES SHOULD BE COMPLETED PRIOR TO THE FINAL INSPECTION.

   FOR RAISED PRESBY SYSTEMS:
   - A. THE MOUND SAND AND SYSTEM SAND MUST BE INSTALLED IN THE MOUND, INCLUDING THE 6" DEPTH OF SYSTEM SAND OVER THE TOPS OF THE PRESBY PIPES. LEAVE ONLY THE TOP OF END CAPS AND CONNECTING PIPES VISIBLE FOR INSPECTION.
   - B. ALL FILL MUST BE PLACED IN THE FILL SLOPES, INCLUDING ALL SAND, COMMON FILL AND TOPSOIL. CONTRACTOR MUST LEAVE THE TOP OF THE SYSTEM SAND UNCOVERED FOR INSPECTION. UPGRADIENT SURFACE DRAINAGE MUST BE INSTALLED.
   - C. ALL VENT LINES MUST BE INSTALLED.
   - D. THE CONTRACTOR MUST HAVE SIEVE ANALYSIS REPORTS FOR THE MOUND SAND AND THE SYSTEM SAND ON HAND AT THE TIME OF THE INSPECTION. THE SIEVE ANALYSIS REPORT MUST IDENTIFY THE SUPPLIER, AND THE DATE OF THE ANALYSIS.
5) ALL OTHER SITE SPECIFIC REQUIREMENTS IDENTIFIED ON THE DRAWING MUST BE COMPLETED, INCLUDING (IF APPLICABLE) ABANDONMENT OF EXISTING FAILED SYSTEMS, ABANDONMENT OF EXISTING WELLS, INSTALLATION OF CURTAIN DRAINS.
6) IF DEFICIENCIES ARE FOUND, THESE WILL BE NOTED AND THE INSPECTION WILL BE CONSIDERED A PARTIAL INSPECTION. A SUBSEQUENT INSPECTION(S) WILL BE REQUIRED TO CONFIRM THE DEFICIENCIES HAVE BEEN CORRECTED. ALL COSTS FOR SUBSEQUENT INSPECTIONS WILL BE CHARGED DIRECTLY TO THE CLIENT.

CONSTRUCTION NOTES:
1) PRIOR TO STARTING CONSTRUCTION, CONTRACTOR SHALL:
   - a. ARRANGE TO HAVE A PRECONSTUCTION MEETING AND/OR CALL WITH THE ENGINEER TO REVIEW THE PROJECT.
   - b. CONFIRM THAT THE CONTRACTOR IS WORKING OFF THE LATEST STATE APPROVED PLAN AND LATEST STATE PERMITS.
   - c. CONFIRM THAT THE CONTRACTOR HAS REVIEWED ALL OTHER PERMITS ASSOCIATED WITH THE CONSTRUCTION.
   - d. CONFIRM THAT THE CONTRACTOR HAS PERFORMED REVIEW OF THE FIELD LAYOUT OF THE LOCATIONS AND ELEVATIONS OF EXISTING AND PROPOSED COMPONENTS AND INFORM THE ENGINEER OF ANY CONFLICTS PRIOR TO CONSTRUCTION.
   - e. FOR REPLACEMENT SEPTIC SYSTEMS, THAT THE CONTRACTOR HAS EXPOSED ALL CONNECTIONS TO EXISTING SEWER COMPONENTS, AND RUN WATER FROM ALL FIXTURES IN THE HOUSE TO CONFIRM THERE ARE NO OTHER EXISTING SEWER CONNECTIONS, PRIOR TO SETTING TANSIC AND CONSTRUCTING THE FIELD.
   - f. PROVIDE FOR ENGINEER'S APPROVAL A COPY OF THE SIEVE ANALYSIS REPRESENTING THE MOUND SAND PROPOSED FOR USE IN THE SYSTEM PRIOR TO COMMENCEMENT OF WORK FOR ENGINEER'S APPROVAL. NO SIEVE ANALYSIS OLDER THAN 1 YEAR WILL BE ACCEPTED. ENGINEER RESERVES THE RIGHT TO DO AN IN PLACE SAMPLE OF THE MOUND SAND AT THE TIME OF THE FINAL INSPECTION TO CONFIRM THE SIEVE ANALYSIS ACCEPTED BY THE ENGINEER IS REPRESENTATIVE OF THE SAND USED TO CONSTRUCT THE MOUND. IT IS THE CONTRACTORS REPSPONSIBILITY TO INSTALL SAND THAT MEETS THE PROJECT SPECIFICATIONS.
2) THE INFILTRATOR SYSTEM SHALL BE INSTALLED IN ACCORDANCE WITH THE APPROVED PLANS AND DESIGN AND INSTALLATION MANUAL FOR INFILTRATOR CHAMBERS IN VERMONT.
4) IN THE AREA OF THE WASTEWATER SYSTEM CUT ALL SITE VEGETATION FLUSH TO GROUND, PLOW AREA SHOWN TO 8" DEPTH, WITHIN LIMITS OF FILL. PLOW PARALLEL WITH CONTOURS, CASTING SOIL UPSLOPE. TILLING OF GROUND WITH BUCKET TEETH IS ACCEPTABLE, IF DONE IN THE SAME MANNER.
5) PLOWING OR CONSTRUCTION OF THE MOUND AREA MAY NOT TAKE PLACE IN HIGH MOISTURE CONTENT CONDITIONS.
6) NO EQUIPMENT SHALL TRACK ON PLOWED SURFACE UNTIL 12" OF SAND FILL HAS BEEN PLACED OVER PLOWED SURFACE.
7) ALL TREES WITHIN 10' OF TOE OF FILL MUST BE CUTOFF AT THE BASE EXCEPT WHERE NOTED, EXERCISE CARE TO MINIMIZE DISTURBANCE OF THE SOIL LAYERS.
8) MOUND SAND FILL MUST MEET ANY OF THE THREE VT MOUND SAND SPECS SHOWN ON PLAN.
9) FROM HOUSE TO SEPTIC TANK USE 4" SCH. 40 PVC BUILDING SEWER, FROM SEPTIC TANK TO DISTRIBUTION BOX USE 4" SCH. 40 PVC OR 4" SDR 35 PVC, FROM DISTRIBUTION BOX TO MULTI-PORT END CAPS AT TRENCHES USE 4" SDR 35 PVC. FOLLOW INVERT ELEVATIONS, MINIMUM AND MAXIMUM SLOPES ON PLANS, FOLLOW DETAILS FOR ALL INSTALLATIONS. PVC JOINTS TO BE SOLVENT WELDED OR GASKETED PUSH-ON TYPE.
10) INFILTRATOR TRENCH SYSTEM IS DESIGNED TO BE VENTED FROM THE FINAL VENT THROUGH THE ROOF VENT OF THE HOUSE. ALL 'TRAPS' OR OBSTRUCTIONS THAT WOULD HINDER VENTING OF THE LEACHING SYSTEM MUST BE AVOIDED. TRENCHES ARE TO BE VENTED THROUGH THE INLET OF MULTI-PORT END CAP AS SHOWN IN DETAIL USING 4" SCH. 40 PVC.
11) COVER OVER RAISED SYSTEM LINE TO BE 2" MINUS MATERIAL, FREE OF ALL LARGE STONES.
12) AT LEAST 72 HOURS BEFORE PLACING SAND FILL OVER PLOWED AREA, NOTIFY ENGINEER FOR INSPECTION. AT LEAST 72 HOURS BEFORE FINAL INSPECTION OF ENTIRE SYSTEM, CONTACT ENGINEER AND SEPTIC OFFICER FOR INSPECTION. CONTRACTOR TO SUPPLY ALL WATER IF REQUIRED.
13) LOAM, SEED & MULCH ALL DISTURBED AREAS. PROVIDE EROSION CONTROL AS REQUIRED. FOLLOW EROSION CONTROL PLAN IF ONE EXISTS FOR PROJECT.
14) CERTIFICATION FORM, PROVIDED BY ENGINEER, STATING THAT SYSTEM WAS INSTALLED ACCORDING TO PLANS (AND OTHER DIRECTIVE FROM ENGINEER) WILL BE COMPLETED BY THE ENGINEER. CONSTRUCTION NOT IN ACCORDANCE WITH PLANS OR ENGINEER'S DIRECTIVES WILL NOT BE APPROVED, AND COULD RESULT IN REMOVAL AND REINSTALLATION TO MEET PROJECT PLANS AND DIRECTIVES.
15) CONTACT ENGINEER WITH ANY CONFLICTS AS SOON AS THEY ARE EVIDENT, PRIOR TO PROCEEDING WITH CONSTRUCTION. ANY CHANGES MUST BE APPROVED BY ENGINEER IN WRITING.
16) LOCATION OF DISPOSAL SYSTEM WILL BE STAKED AND LABELED IN THE FIELD. CONTRACTOR TO CONFIRM THAT THE LOCATIONS OF STAKES ARE ACCURATE AND IMMEDIATELY REPORT ANY CONFLICTS TO ENGINEER PRIOR TO COMMENCING CONSTRUCTION. NO UNNECCESSARY DISTURBANCE OR STORAGE OF MATERIALS IS ALLOWED IN THE DISPOSAL FIELD AREA. IT IS THE CONTRACTOR'S RESPONSIBILITY TO MAINTAIN THE STAKES, AND TEMPORARY BENCHMARK IN THEIR ORIGINAL POSITIONS AS STAKED OUT BY THE ENGINEER.
17) THE CONTRACTOR SHALL PROVIDE THE ENGINEER WITH A COPY OF THE SIEVE ANALYSIS REPRESENTING THE MOUND SAND PROPOSED FOR USE IN THE SYSTEM CONSTRUCTION PRIOR TO COMMENCEMENT OF WORK FOR ENGINEER'S APPROVAL.
18) OWNER AND CONTRACTOR ARE REMINDED THAT PROJECT IS NOT COMPLETE UNTIL AN INSTALLATION CERTIFICATION LETTER IS OBTAINED. TRINITY CANNOT ISSUE THE INSTALLATION CERTIFICATION LETTER UNTIL THE WASTEWATER SYSTEM AND PROPOSED WELL ARE CONSTRUCTED IN ACCORDANCE WITH THE PLAN APPROVED BY THE VT DEC.

MOUND SAND FILL MUST MEET ONE OF THESE SPECIFICATIONS		
TYPE 1	TYPE 2 (ASTM C-33)	TYPE 3
NO. 3/8 85-100% NO. 40 25-75% NO. 60 0-30% NO. 100 0-10% NO. 200 0-5%	NO. 4 95-100% NO. 8 80-100% NO. 16 50-85% NO. 30 25-60% NO. 50 10-30% NO. 100 2-10%	NO. 3/8 85-100% NO. 40 30-50% NO. 200 0-5%
A SIEVE ANALYSIS OF PROPOSED SAND SOURCE MUST BE SUBMITTED PRIOR TO CONSTUCTION, FOR APPROVAL. NO LIMESTONE BASED SAND ALLOWED.		

DESIGN MEETS THE MINIMUM REQUIREMENTS OF VT DEC CHAPTER 1 RULES, NOVEMBER 6, 2023.

APPLICATION RATE BASAL AREA — 0.74 GAL/SF/DAY (BASAL APPLICATION RATE)

DESIGN FLOWRATE BASED ON
- 3 BEDROOM AT @ 140 = 420 GAL/DAY
- 1 BEDROOM AT @ 70 = 70 GAL/DAY
- 4 TOTAL BEDROOMS — TOTAL DESIGN FLOW = 490 GAL/DAY

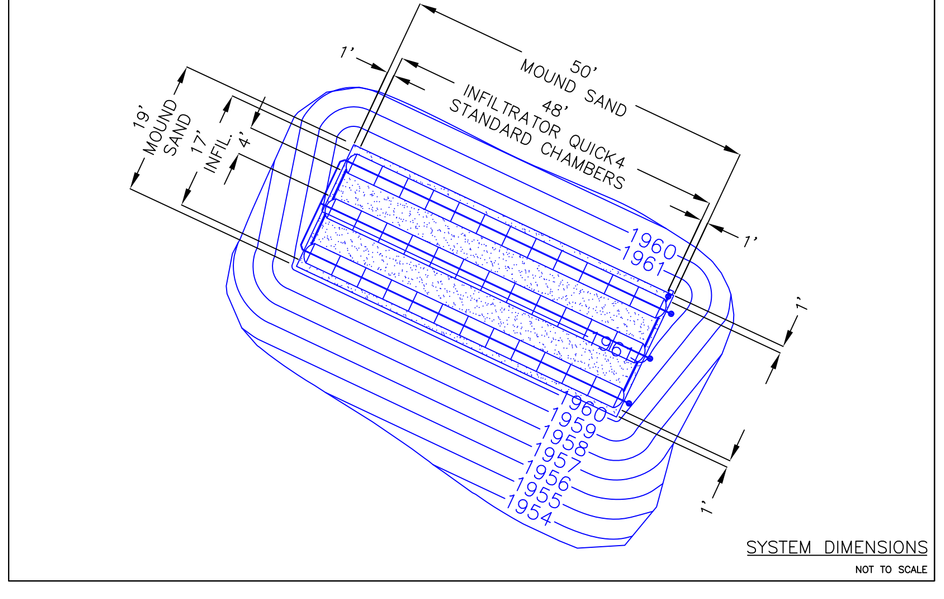
MIN. BASAL AREA REQUIRED = (490 GAL/DAY) / (0.74 GAL/SF/DAY) = 662 SF
APPROXIMATE EFFECTIVE BASAL AREA PROVIDED = 1768 SF F.S. ON AREA 2.67

SEPTIC TANK SIZE TO BE MINIMUM OF 1500 GALLONS, USE EFFLUENT TEE FILTER.

NUMBER OF QUICK4 STANDARD CHAMBERS REQUIRED FROM INFILTRATOR LEACHING CHAMBERS VERMONT STATE ATTACHMENT = 35 Q4 STANDARD CHAMBERS
TOTAL NUMBER OF QUICK4 CHAMBERS PROVIDED = 36 Q4 STANDARD CHAMBERS
FACTOR OF SAFETY ON CHAMBERS = 1.03

LINEAR LOADING RATE (LLR): LLR = (h)(f)
SITE SLOPE = 9.0%
SOIL TYPE = Sandy Loam
h = 19 " (Critical Depth) - 6 " (Min. Unsaturated Soil Required) — 13.0 Inches (Max. GW Mounding Allowed)
f = 33.7 (From VTDEC Simplified Desktop Mounding Analysis)
LLR (Max.) = 36.51 gallons/day/Lineal Foot (Would produce Max. allowable GW Mounding = 13.0 ")
LLR (actual) = 10.21 gallons/day/Lineal Foot (LLR actual, Actual resulting GW Mounding = 3.6 ")

OTHER DESIGN NOTES:
1) NO GARBAGE DISPOSAL SHALL BE USED, NO WATER SOFTNER SHALL BE INSTALLED.
2) GROUNDWATER FOUND AT 19 THEREFORE PERFORMANCE BASED APPROACH IS USED.
3) SYSTEM MUST BE INSTALLED AND OPERATED IN ACCORDANCE WITH THE INFILTRATOR DESIGN AND INSTALLATION MANUAL, & THE VT STATE ATTACHMENT.
4) DESIGN IS BASED ON A MINIMUM OF 21 INCHES OF SAND UNDER ARC CHAMBERS.
5) SYSTEM IS DESIGNED TO TREAT WASTEWATER EFFLUENT FROM A RESIDENTIAL HOME. THE ADDITION OF ABUSIVE SUBSTANCES, ADDITIVES AND CONSTANT DISCHARGE OF WATER ARE DETRIMENTAL TO THE SYSTEM.



SYSTEM DIMENSIONS

NOT TO SCALE

TRINITY ENGINEERING
TECHNICAL SERVICES, LLC
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REV. NO.	DATE	DESCRIPTION	BY
2	7/16/26	AMEND LOT 2 TO GRAVITY FLOW	J.E.D.
1	3/28/25	ORIGINAL SUBMISSION FOR PERMIT	J.E.D.

STATE OF VERMONT CIVIL JOHN E. DUPRAS No. 7376 LICENSED PROFESSIONAL ENGINEER

DESIGNED BY: J.E.D.
DRAWN BY: G.J.V.
CHECKED BY: J.E.D.
SCALE: AS SHOWN

TITLE: WASTEWATER DISPOSAL SYSTEM AND POTABLE WATER SUPPLY DESIGN

PROJECT: PREPARED FOR DAVID OSTAFIN
PROPOSED LOT 2, HIGLEY HILL ROAD
TOWN OF DOVER, VERMONT

DATE: 7/16/2026

PROJECT NUMBER: 26096

SHEET NUMBER: C-2

SHEET: 2 OF 2